

ALGEBRA CHEATSHEET

Everything you need for secondary school: from basic rules to quadratics, functions and logs

$$x^2 + y^2 = r^2$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

1 Order of Operations & Laws

BIDMAS: Brackets → Indices → Div/Mult
→ Add/Sub (left to right)

$$a + b = b + a \quad ab = ba \quad (\text{commutative})$$

$$(a + b) + c = a + (b + c) \quad (\text{associative})$$

$$a(b + c) = ab + ac \quad (\text{distributive})$$

Signs: $(-)(-) = +$ $(-)(+) = -$ $-(-a) = +a$

Like terms have identical letters & powers:

$$3x^2y + 5x^2y = 8x^2y$$

Watch out: $(a + b)^2 \neq a^2 + b^2$ $\sqrt{a + b} \neq \sqrt{a} + \sqrt{b}$

2 Laws of Indices

$$a^m \times a^n = a^{m+n} \quad a^m \div a^n = a^{m-n}$$

$$(a^m)^n = a^{mn} \quad (ab)^n = a^n b^n$$

$$a^0 = 1 \quad a^1 = a$$

$$a^{-n} = \frac{1}{a^n} \quad \left(\frac{a}{b}\right)^{-n} = \left(\frac{b}{a}\right)^n$$

$$a^{\frac{1}{n}} = \sqrt[n]{a} \quad a^{\frac{m}{n}} = (\sqrt[n]{a})^m$$

Same base needed to add/subtract powers. Coefficients multiply separately: $3x^2 \cdot 4x^5 = 12x^7$.

3 Surds

$$\sqrt{a}\sqrt{b} = \sqrt{ab} \quad \frac{\sqrt{a}}{\sqrt{b}} = \sqrt{\frac{a}{b}} \quad (\sqrt{a})^2 = a$$

$$\text{Simplify: } \sqrt{50} = \sqrt{25 \cdot 2} = 5\sqrt{2}$$

Rationalise: $\frac{1}{\sqrt{a}} = \frac{\sqrt{a}}{a}$ $\frac{1}{a + \sqrt{b}} = \frac{a - \sqrt{b}}{a^2 - b}$
(multiply by conjugate)

4 Expanding Brackets

$$(a + b)(c + d) = ac + ad + bc + bd \quad (\text{FOIL})$$

$$(a + b)^2 = a^2 + 2ab + b^2$$

$$(a - b)^2 = a^2 - 2ab + b^2$$

$$(a + b)(a - b) = a^2 - b^2$$

$$(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

$$\text{Ex: } (x + 3)(x - 5) = x^2 - 2x - 15$$

5 Factorising

1. Common factor: $6x^2 + 9x = 3x(2x + 3)$

2. Difference of two squares: $a^2 - b^2 = (a + b)(a - b)$

3. Quadratic $x^2 + bx + c = (x + p)(x + q)$
where $pq = c$, $p + q = b$

4. $ax^2 + bx + c$: find p, q with $pq = ac$, $p + q = b$;
split the middle term, then group:

$$2x^2 + 7x + 3 = 2x^2 + 6x + x + 3 = (2x + 1)(x + 3)$$

5. Grouping: $ax + ay + bx + by = (a + b)(x + y)$

Cubes: $a^3 \pm b^3 = (a \pm b)(a^2 \mp ab + b^2)$

Always take out the HCF first!

6 Equations & Inequalities

Golden rule: do the same to both sides.
Undo operations in reverse BIDMAS.

$$3(x - 2) = 2x + 5 \Rightarrow 3x - 6 = 2x + 5 \Rightarrow x = 11$$

Inequalities behave the same, **except:** multiplying or dividing by a **negative flips** the sign.

$$-2x < 6 \Rightarrow x > -3$$

Number line: $\circ$ open for $<, >$ $\bullet$ closed for $\leq, \geq$

$$|x| < a \Leftrightarrow -a < x < a \quad |x| > a \Leftrightarrow x < -a \text{ or } x > a$$

7 Simultaneous Equations

Elimination: match coefficients, then add or subtract.

Substitution: rearrange one equation for y , substitute into the other.

$$\begin{cases} x + y = 7 \\ x - y = 1 \end{cases} \Rightarrow 2x = 8, x = 4, y = 3$$

Linear + quadratic: substitute the line into the curve $\Rightarrow$ solve the quadratic (0, 1 or 2 intersections).

8 Quadratics $ax^2 + bx + c = 0$

Factorise: $(x - p)(x - q) = 0 \Rightarrow x = p$ or q

Formula: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Discriminant $\Delta = b^2 - 4ac$:

$\Delta > 0$: 2 real roots $\Delta = 0$: 1 repeated root

$\Delta < 0$: no real roots

Completing the square:

$$x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 - \left(\frac{b}{2}\right)^2 + c$$

$$a(x + h)^2 + k \Rightarrow \text{vertex } (-h, k), \text{ axis } x = -h$$

Roots: sum $= -\frac{b}{a}$ product $= \frac{c}{a}$

Parabola: $a > 0 \cup$ (min) $a < 0 \cap$ (max)

Vertex at $x = -\frac{b}{2a}$ y -intercept $= c$

9 Algebraic Fractions

Simplify: factorise top & bottom, then cancel factors.

$$\frac{x^2 - 9}{x^2 + x - 12} = \frac{(x + 3)(x - 3)}{(x + 4)(x - 3)} = \frac{x + 3}{x + 4}$$

$$\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd} \quad \frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$$

$$\frac{a}{b} \pm \frac{c}{d} = \frac{ad \pm bc}{bd} \quad (\text{common denominator})$$

Never cancel terms: $\frac{x + 2}{2} \neq x + 1$; divide every term: $\frac{x}{2} + 1$.

10 Straight Lines

$$y = mx + c \quad m = \text{gradient}, c = y\text{-intercept}$$

$$\text{Gradient } m = \frac{y_2 - y_1}{x_2 - x_1} \quad y - y_1 = m(x - x_1)$$

$$\text{Midpoint } \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

$$\text{Distance } d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Parallel: $m_1 = m_2$ **Perpendicular:**

$$m_1 m_2 = -1$$

Circle: $(x - a)^2 + (y - b)^2 = r^2$ (centre (a, b))

11 Sequences & Series

Arithmetic (add d): $u_n = a + (n - 1)d$

$$S_n = \frac{n}{2} [2a + (n - 1)d] = \frac{n}{2} (a + l)$$

Geometric (multiply by r): $u_n = ar^{n-1}$

$$S_n = \frac{a(1 - r^n)}{1 - r} \quad S_\infty = \frac{a}{1 - r}, |r| < 1$$

Quadratic n th term: 2nd difference $= 2a$, so $an^2 + bn + c$

Sums: $\sum n = \frac{n(n + 1)}{2}$

$$\sum n^2 = \frac{n(n + 1)(2n + 1)}{6}$$

12 Changing the Subject

Treat the formula like an equation: *undo* operations around the target letter.

$$v^2 = u^2 + 2as \Rightarrow s = \frac{v^2 - u^2}{2a}$$

Subject in two places? Collect those terms on one side, factorise, then divide:

$$ax + b = cx + d \Rightarrow x(a - c) = d - b \Rightarrow x = \frac{d - b}{a - c}$$

13 Direct & Inverse Proportion

Direct: $y \propto x \Rightarrow y = kx$ $y \propto x^2 \Rightarrow y = kx^2$

Inverse: $y \propto \frac{1}{x} \Rightarrow y = \frac{k}{x}$

Find k from one pair of values, write the formula, then use it.

Exponential growth/decay: $y = ab^x$ ($b > 1$ grow, $0 < b < 1$ decay)

14 Functions

Composite: $fg(x) = f(g(x))$ (do g first)

Inverse: $y = f(x)$, swap $x \leftrightarrow y$, solve for $y \Rightarrow f^{-1}(x)$

$$ff^{-1}(x) = x \quad \text{graph reflects in } y = x$$

Domain = allowed inputs **Range** = outputs

Domain limits: $\frac{1}{x}$ ($x \neq 0$), $\sqrt{x}$ ($x \geq 0$)

15 Graph Transformations

$$y = f(x) + a \text{ up } a$$

$$y = f(x + a) \text{ left } a \text{ (opposite!)}$$

$$y = af(x) \text{ stretch } \uparrow \text{ by factor } a$$

$$y = f(ax) \text{ stretch } \leftrightarrow \text{ by factor } \frac{1}{a}$$

$$y = -f(x) \text{ reflect in } x\text{-axis}$$

$$y = f(-x) \text{ reflect in } y\text{-axis}$$

16 Logarithms

$$y = a^x \Leftrightarrow x = \log_a y$$

$$\log_a a = 1 \quad \log_a 1 = 0$$

$$\log a + \log b = \log ab \quad \log a - \log b = \log \frac{a}{b}$$

$$n \log a = \log a^n \quad \log_a b = \frac{\log_c b}{\log_c a}$$

$$\text{Solve } a^x = b \Rightarrow x = \frac{\log b}{\log a} \quad \ln x = \log_e x$$

17 Polynomials

Remainder theorem: $f(x) \div (x - a)$ leaves remainder $f(a)$.

Factor theorem: $f(a) = 0 \Leftrightarrow (x - a)$ is a factor.

Method: test $\pm$ factors of the constant, divide, factorise the rest.

Degree n polynomial has at most n real roots.

18 Know Your Graphs

$$y = mx + c \text{ straight line}$$

$$y = x^2 \text{ U-shaped parabola}$$

$$y = x^3 \text{ S-shape through origin}$$

$$y = \frac{1}{x} \text{ hyperbola, 2 branches}$$

$$y = a^x \text{ exponential, through } (0, 1)$$

$$x^2 + y^2 = r^2 \text{ circle at origin}$$

Roots = where the graph meets the x -axis ($y = 0$).

Common Mistakes & Exam Tips

X $(x + y)^2 \neq x^2 + y^2$ **X** $\frac{a}{b + c} \neq \frac{a}{b} + \frac{a}{c}$

X Forgetting $\pm$ when square-rooting

X Dropping signs: $-(x - 3) = -x + 3$

✓ Check answers by substituting back.

✓ Show every step: method marks matter.

✓ Set one side to 0 before solving quadratics.